



DWF_dp_rndsats functions

Arithmetic Rounding and Saturation

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Description

Revision History

The DWF_dp_rndsats functions truncate the lower bits of argument *a* below the bit position specified by argument *lsb*, round according to the rounding mode specified by argument *mode*, truncate the upper bits above the bit position specified by argument *msb* and return a saturated value if an overflow (or underflow) occurs. Argument *a* and the return value are both either signed (two's complement) or unsigned.

Table 1-1 Function Names

Function Name	Description
DWF_dp_rndsats	VHDL unsigned rounding and saturation
DWF_dp_rndsats	VHDL signed (two's complement) rounding and saturation
DWF_dp_rndsats_uns	Verilog unsigned rounding and saturation
DWF_dp_rndsats_tc	Verilog signed (two's complement) rounding and saturation

Table 1-2 Argument Description

Argument Name	Type	Width / Values	Description
<i>a</i>	Vector	width	Input data
<i>msb</i>	Integer	$> lsb, \leq width-1$	MSB index of return value (VHDL only, constant)
<i>lsb</i>	Integer	$\geq 1, < msb$	LSB index of return value (VHDL only, constant)
<i>mode</i>	Vector	4	Rounding mode Find more information on rounding and rounding modes in the DWF_dp_rnd Datasheet .
DWF_dp_rndsats	Vector	<i>msb</i> – <i>lsb</i> +1	Return value

Table 1-3 Parameter Description (Verilog)

Parameter	Values	Description
<i>width</i>	≥ 1	Word length of input <i>a</i>
<i>msb</i>	$> lsb, \leq width - 1$	MSB index of return value
<i>lsb</i>	$\geq 1, < msb$	LSB index of return value

Verilog Include File: DW_dp_rndsats_function.inc

Functional Description

DWF_dp_rndsats is functionally equivalent to rounding followed by saturation (also requires special handling of rounding overflow).

```
z[msb-lsb:0] = DWF_dp_rndsats (a[width-1:0], msb, lsb, mode)
z[msb-lsb:0] = DWF_dp_sat (
    DWF_dp_rnd (a[width-1:0], lsb, mode),
    msb-lsb+1)
```

Unsigned / Signed Rounding

```
t[width-lsb-1:0] = a[width-1:lsb]+1 if a[lsb-1:0] > threshold(mode)
                  = a[width-1:lsb]   else
```

Find more information on rounding and rounding modes in the [DWF_dp_rnd Datasheet](#).

Unsigned Saturation

```
z[msb-lsb:0] = 2msb-lsb+1-1 if t[width-lsb-1:0] > 2msb-lsb+1-1
              = t[msb-lsb:0] else
```

Signed Saturation

```
z[msb-lsb:0] = 2msb-lsb-1 if t[width-lsb-1:0] > 2msb-lsb-1
              = -2msb-lsb else if t[width-lsb-1:0] < -2msb-lsb
              = t[msb-lsb:0] else
```

For more information about the DesignWare datapath functions, refer to the topic titled [DesignWare Datapath Functions Overview](#).

Related Topics

- [DesignWare Datapath Functions Overview](#)
- [DesignWare Building Block IP User Guide](#)

VHDL Example

```
library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use IEEE.numeric_std.all;
use DWARE.DW_dp_functions.all;
-- DWARE.DW_dp_functions_arith package if IEEE.std_logic_arith is used

entity DWF_dp_rndsats_test is
  port (a, b, c : in  signed(7 downto 0);
        z       : out signed(7 downto 0));
end DWF_dp_rndsats_test;

architecture rtl of DWF_dp_rndsats_test is
begin
  z <= DWF_dp_rndsats (a * b, 11, 4, DW_dp_rnd_near_even) + c;
end rtl;
```

Verilog Example

```
module DWF_dp_rndsats_test (a, b, c, z);

    input  signed [7:0] a, b, c;
    output signed [7:0] z;

    // Passes the parameters to the function
    parameter width = 16;
    parameter msb   = 11;
    parameter lsb   = 4;

    // add "$SYNOPSISYS/dw/sim_ver" to the search path for simulation
    `include "DW_dp_rndsats_function.inc"

    assign z = DWF_dp_rndsats_tc (a * b, DW_dp_rnd_near_even) + c;

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes ■ Added this Revision History table and the document links on this page

Copyright Notice and Proprietary Information

© 2024 Synopsys, Inc. All rights reserved. This Synopsys software and all associated documentation are proprietary to Synopsys, Inc. and may only be used pursuant to the terms and conditions of a written license agreement with Synopsys, Inc. All other use, reproduction, modification, or distribution of the Synopsys software or the associated documentation is strictly prohibited.

Destination Control Statement

All technical data contained in this publication is subject to the export control laws of the United States of America. Disclosure to nationals of other countries contrary to United States law is prohibited. It is the reader's responsibility to determine the applicable regulations and to comply with them.

Disclaimer

SYNOPSYS, INC., AND ITS LICENSORS MAKE NO WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, WITH REGARD TO THIS MATERIAL, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

Trademarks

Synopsys and certain Synopsys product names are trademarks of Synopsys, as set forth at <https://www.synopsys.com/company/legal/trademarks-brands.html>.

All other product or company names may be trademarks of their respective owners.

Free and Open-Source Software Licensing Notices

If applicable, Free and Open-Source Software (FOSS) licensing notices are available in the product installation.

Third-Party Links

Any links to third-party websites included in this document are for your convenience only. Synopsys does not endorse and is not responsible for such websites and their practices, including privacy practices, availability, and content.

Synopsys, Inc.
www.synopsys.com